



DIGITAL ECONOMY OUTLOOK 2026

The economics of AI adoption in Nigeria

Banking • Energy • Telecommunications • Manufacturing

FROM PILOTS TO PRODUCTIVITY

A decision-led view of where artificial intelligence can create economic value—and what prevents that value from being realised.

4

SECTORS

one operating model

DESK-RESEARCH AND MODEL-BASED EDITION • DATA CUT-OFF 25 AUGUST 2026

READER NOTE

What this report measures—and what it does not

BOTTOM LINE

Nigeria has no harmonised public dataset for AI expenditure, deployment depth or realised return across these four sectors. This report therefore combines verified operating indicators with explicitly labelled Renalytica analyst assessments. It does not present a national AI market-size estimate.

The economic unit of analysis

The report evaluates the conversion of artificial intelligence into operating value. AI adoption is treated as an investment programme—not a software licence. The relevant return is the incremental revenue, cost productivity, loss avoidance and asset or yield improvement left after data, integration, compute, talent, change and control costs.

CORE EQUATION

Realised annual value = addressable operating base × achievable improvement × adoption depth × implementation reliability – recurring AI cost – control and remediation cost.

Evidence convention

Label	Meaning	How to read it
Observed	Published by a regulator, company or statistics authority	Context, not automatic evidence of AI causality
Company-reported	A named organisation’s disclosure	Case evidence; not representative of the whole sector
Analyst assessment	Renalytica scoring or band based on the disclosed method	Directional decision support; not a forecast
Illustrative model	A transparent scenario or indexed example	Tests economics under stated assumptions

Scope: Nigerian banking, telecommunications, energy and manufacturing. Currency values are avoided where evidence is not comparable. Data cut-off: 25 August 2026.

CONTENTS

A practical route from evidence to action

Section	Pages	Decision question
Executive view	4–6	Where is value most likely to be realised?
Sector economics	8–15	Which use cases fit each operating model?
Scaling the investment	17–20	What does it cost, and how should it be governed?
Outlook and action	22–24	What should leaders do now?

How to use the report

FOR BOARDS AND INVESTORS

Start with pages 4–6, then use the sector page and cost stack to challenge whether announced AI programmes can convert into measurable cash and risk outcomes.

FOR OPERATORS AND POLICYMAKERS

Use the use-case tables, governance gates and workforce page to distinguish scalable infrastructure from isolated demonstrations.

Four questions run through every page

- Is the decision frequent, measurable and supported by accessible data?
- Can the model be embedded in a workflow without creating an unowned control gap?
- Does value survive integration, change and ongoing monitoring costs?
- Can the organisation stop, override or explain the outcome when the model is wrong?

DESIGN NOTE

The report adopts the disciplined hierarchy, section rhythm and data-led readability associated with leading professional-services publications, interpreted through Renalytica’s grey-and-orange identity. All analysis and visual construction are original.

EXECUTIVE VIEW

Nigeria’s AI prize is operating leverage—not novelty

The four sectors share model access, but their economics diverge because the underlying decisions, assets and control burdens differ.



The headline

Banking and telecommunications have the clearest near-term route to scale because they combine high-frequency decisions, large digital datasets and distribution across millions of customer interactions. Energy can generate high value from uptime, loss reduction and maintenance, but integration with operational technology and the cost of failure slow deployment. Manufacturing presents the widest diffusion gap: the value case is strongest at a specific line, machine or quality gate, while plant-wide transformation requires better instrumentation and data discipline.

Five findings

	Finding	Implication
01	The model is not the business case	Access to a capable model is increasingly commoditised. Value comes from proprietary data, workflow redesign, adoption and controls.
02	Start where errors are reversible	High-volume, measurable, human-overridable decisions should precede safety-critical or fully autonomous use.
03	Integration dominates cost	Data engineering, workflow integration and change typically exceed the direct cost of software or model access.
04	Loss avoidance matters	Fraud, outage, downtime, leakage, defects and energy waste can be economically larger than labour savings.
05	Governance is a scaling mechanism	Clear ownership, evaluation and override rules shorten the distance between a promising pilot and an auditable operating capability.

EXECUTIVE VIEW

A sector scorecard for investment sequencing

Scores are 1–5 Renalytica analyst assessments. A high control-burden score means more—not less—governance is required.

AI economics differ materially by sector					
Banking	4.3	4.6	4.5	4.5	4.8
Telecom	4.5	4.5	4.3	4.7	4.0
Energy	3.1	4.4	3.2	3.2	4.6
Manufacturing	2.8	3.8	3.4	3.0	3.5
	Data readiness	Value density	Speed to value	Scale repeatability	Control burden

Exhibit 1. Sector AI economics scorecard

Source: Renalytica analyst assessment. Scoring reflects public evidence on digital operations, use-case repeatability, asset economics and regulatory exposure..

SCALE FIRST

Banking and telecom can scale assistive and decision AI rapidly—provided production deployment is tied to loss, service, revenue or network metrics rather than model demonstrations.

INSTRUMENT FIRST

Energy and manufacturing often need better sensor data, asset taxonomy, maintenance history and integration before advanced AI can outperform simpler analytics.

INVESTOR LENS

A high-readiness sector can still destroy value through poorly governed scale. A lower-readiness sector can create attractive returns with one tightly bounded, high-value asset use case.

EXECUTIVE VIEW

The value pool is four-dimensional

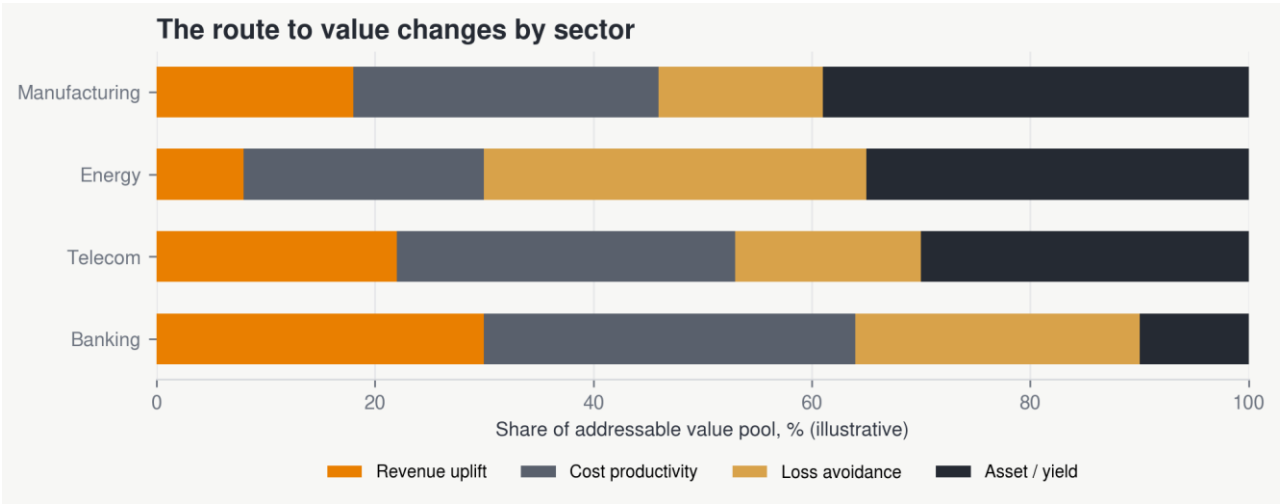


Exhibit 2. Illustrative mix of addressable AI value levers by sector

Source: Renalytica illustrative model. Shares indicate relative emphasis, not measured national value pools.

Value lever	Typical metrics	Failure mode
Revenue uplift	conversion, retention, price realisation, collections	attribution confuses growth with model effect
Cost productivity	cost per interaction, cycle time, rework, throughput	time saved does not become capacity
Loss avoidance	fraud, leakage, outage, defect, safety incidents	avoided loss lacks a credible counterfactual
Asset / yield	availability, energy intensity, yield, utilisation	model is not integrated into maintenance or control

MEASUREMENT RULE

Approve a production AI use case only when the operating baseline, decision owner, intervention path and value-realisation mechanism are visible before the pilot starts.

01

Sector economics

Four sectors. Four different routes from model capability to realised value.

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SECTOR ECONOMICS / BANKING

The strongest near-term monetisation—and the highest conduct risk



Banks possess transactional data, digital distribution and repeated decisions across service, fraud, credit, collections and compliance. That gives AI a dense operating surface. The constraint is not identifying use cases; it is proving causality, protecting customers and keeping human accountability where outcomes affect access to finance.

Best economic wedges

Use case	Value mechanism	Primary measure	Control requirement
Service copilot	more capacity; lower handling time	cost/contact; resolution	grounding, escalation, disclosure
Fraud / AML	loss avoidance; analyst focus	loss rate; alert precision	explainability; case audit trail
Credit / collections	risk-adjusted growth	approval, loss, cure rate	fairness; human review
Operations	cycle-time and error reduction	straight-through rate	access and change controls

PUBLIC EVIDENCE

UBA reported 5.8 million users for its Leo chatbot and 107.8% year-on-year growth in digital banking income. These figures show distribution and channel economics; they do not isolate AI’s causal contribution.

Sources: UBA, 2025 brand release; CBN reform and NeFF pages, accessed 25 August 2026.

BANKING

A bank should fund decisions—not models

01 BASELINE loss, effort or conversion	02 SHADOW compare without acting	03 PILOT bounded customers / teams	04 CONTROL bias, drift, override	05 SCALE embed and monitor
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Illustrative unit economics

EXAMPLE If a service copilot saves two minutes per eligible contact but only half of that time becomes deployable capacity, the value case must use one minute—not two. If adoption reaches only 60% of agents, apply the saving to 60% of eligible volume. Add licence, integration, quality review and escalation costs before calculating payback.		
Decision	Go when...	Stop or redesign when...
Service copilot	quality holds and saved time becomes capacity	agents rework answers or escalations rise
Fraud model	precision improves without missed-loss deterioration	false positives harm genuine customers
Credit model	risk-adjusted margin improves across segments	performance or fairness degrades by cohort
Compliance AI	review effort falls with complete evidence trails	citations are missing or non-reproducible

REGULATORY ECONOMICS CBN’s 2026 emphasis on rapid fraud response, automated AML/CFT standards and enterprise fraud monitoring raises the value of decision speed—but also the cost of weak auditability.
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SECTOR ECONOMICS / TELECOMMUNICATIONS

The best scale economics sit inside the network

179.6m SUBSCRIPTIONS NCC, 2025	82.87% TELEDENSITY NCC, 2025	9 MW DATA-CENTRE PLAN MTN Nigeria disclosed
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Telecommunications pairs enormous event volumes with an expensive physical network. AI can therefore create value at both ends: customer decisions such as churn and service, and asset decisions such as traffic forecasting, energy optimisation and predictive maintenance. A model can scale across millions of interactions, but inference cost, data privacy and production reliability matter at the same scale.

Where economics are strongest

Use case	Value mechanism	Economic test
Network optimisation	capacity deferral; better quality	capex avoided plus churn effect
Energy optimisation	lower tower and data-centre energy	energy saved net of control risk
Predictive maintenance	fewer outages; better field productivity	downtime and truck rolls avoided
Churn / next-best action	retention and wallet growth	incremental margin, not gross revenue
Service automation	lower cost-to-serve	resolution quality net of escalation

INFRASTRUCTURE SIGNAL

MTN Nigeria disclosed a 9 MW data-centre plan, with a first 4.5 MW Tier III phase, while Airtel Africa describes a 38 MW Lagos facility under construction for cloud and AI workloads. Capacity is an enabler; utilisation and customer workloads determine return.

TELECOMMUNICATIONS

Network AI should be valued as avoided constraint

The wrong telecom business case counts every prediction as value. The right one traces a prediction to an operating intervention: a load forecast changes capacity allocation; an anomaly alert prevents an outage; a churn score changes a targeted offer. Value exists only when the intervention changes a measurable outcome.

Layer	AI decision	Value owner	KPI
Radio / core	traffic and fault prediction	CTO / network	availability; congestion; capex
Energy	site load and source optimisation	network operations	kWh/site; fuel; uptime
Commercial	propensity and churn treatment	consumer / enterprise	incremental margin; churn
Care	intent, resolution and routing	customer operations	resolution; repeat contacts
Security	fraud and anomaly detection	risk / security	loss; precision; containment

Investment implication

SHARED PLATFORMS WIN

Feature stores, model monitoring and identity controls can serve network, commercial and service teams. Shared foundations lower marginal use-case cost.

INFERENCE NEEDS A BUDGET

High-volume models can make small per-event costs material. Unit economics should include inference, latency and fall-back capacity at peak load.

PRIORITY

Scale network and service decisions that are frequent, observable and reversible; treat autonomous network control as a later maturity stage with rigorous fall-back engineering.

SECTOR ECONOMICS / ENERGY

Small improvements can carry high value when assets are scarce



Energy economics are dominated by asset availability, loss, throughput and safety. That makes predictive maintenance, load forecasting, anomaly detection and production optimisation attractive. But AI value depends on reliable sensors, maintenance history, operational integration and a safe intervention path—not on a dashboard score.

Use case	Economic value	Deployment constraint
Predictive maintenance	outage and failure avoided	thin failure labels; maintenance adoption
Load forecasting	dispatch and supply planning	weather, demand and network data quality
Loss / meter analytics	commercial and technical loss reduction	meter coverage; field enforcement
Pipeline / site anomaly	production and security protection	false alarms; connectivity; response
Back-office automation	cycle-time and error reduction	process ownership; access controls

EVIDENCE BOUNDARY

NERC’s April 2026 factsheet provides the operating baseline above. It does not attribute availability to AI. NERC has separately discussed AI in regulatory and operational compliance, signalling policy interest rather than measured sector return.

ENERGY

The highest-value model may be the one that stays in shadow mode longest

For critical infrastructure, learning value and operating value must be separated. A model can run in parallel with engineers, build an evidence base and improve alert precision before it is permitted to influence maintenance schedules or operational control. This extends the pilot but reduces the expected cost of failure.

Stage	Permitted role	Evidence required
Observe	describe and classify historic events	data lineage; reproducible back-test
Shadow	predict without changing operations	precision, recall, lead time, false alarms
Recommend	human accepts or rejects intervention	override reasons; realised outcomes
Constrain	automate within hard operating limits	fail-safe logic; cyber and safety test
Operate	closed-loop execution	independent assurance; live monitoring

Economic discipline

- Value avoided downtime using a credible counterfactual—not maximum nameplate output.
- Include sensor retrofit, historian integration, connectivity and maintenance workflow cost.
- Measure alert-to-action time; a correct alert with no field response has no operating value.
- Price cyber, safety and fallback engineering into the investment case from day one.

PUBLIC CASE

NNPC’s Research, Technology and Innovation unit describes robotic process automation for data entry, report generation and system integration. It reports intended efficiency and error-reduction benefits but does not publish a quantified realised return.

SECTOR ECONOMICS / MANUFACTURING

The strongest business case is usually one line, one machine or one quality gate



Manufacturing’s value case is shaped by energy cost, imported inputs, equipment availability, yield, working capital and quality. AI can address each, but sector-wide adoption will be uneven because plant instrumentation, enterprise systems and maintenance data vary materially. A bounded asset use case is more investable than a generic enterprise AI programme.

Use case	Value mechanism	Minimum data
Predictive maintenance	availability and spare-parts productivity	condition, failures, work orders
Vision inspection	scrap, rework and complaints avoided	labelled defect images; line context
Process / yield	more good output per input	recipes, process conditions, quality
Energy optimisation	lower energy per good unit	meter, production and tariff data
Planning / inventory	service level and working capital	orders, stock, lead times, constraints

MACRO SIGNAL

NBS reported manufacturing real growth of 1.41% in 2025 and an 8.05% real-GDP share. Slow sector growth strengthens the productivity imperative, but it does not prove AI adoption or return.

MANUFACTURING

Prove value at the bottleneck, then replicate

01 SELECT costly, frequent bottleneck	02 INSTRUMENT quality and asset data	03 SHADOW compare with operators	04 EMBED work order or control	05 REPLICATE same asset family
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A bankable pilot design

Element	Good design	Weak design
Boundary	one line or asset family	the whole plant
Baseline	yield, downtime or kWh/good unit	anecdotal pain point
Intervention	named operator and workflow	dashboard without action owner
Counterfactual	comparable shifts, products and conditions	before/after without controls
Scale logic	repeatable asset and data template	custom rebuild at every site

BUY WHERE COMMON

Vision platforms, copilots and maintenance tooling can be bought where processes are standard and integration is manageable.

BUILD WHERE PROPRIETARY

Process recipes, failure signatures, quality thresholds and dispatch constraints often contain the differentiated economics.

DECISION RULE

Do not approve the next plant until the first pilot shows realised value after operator adoption, integration, model monitoring and maintenance response cost.

CROSS-SECTOR COMPARISON

Value is not the constraint; conversion capacity is

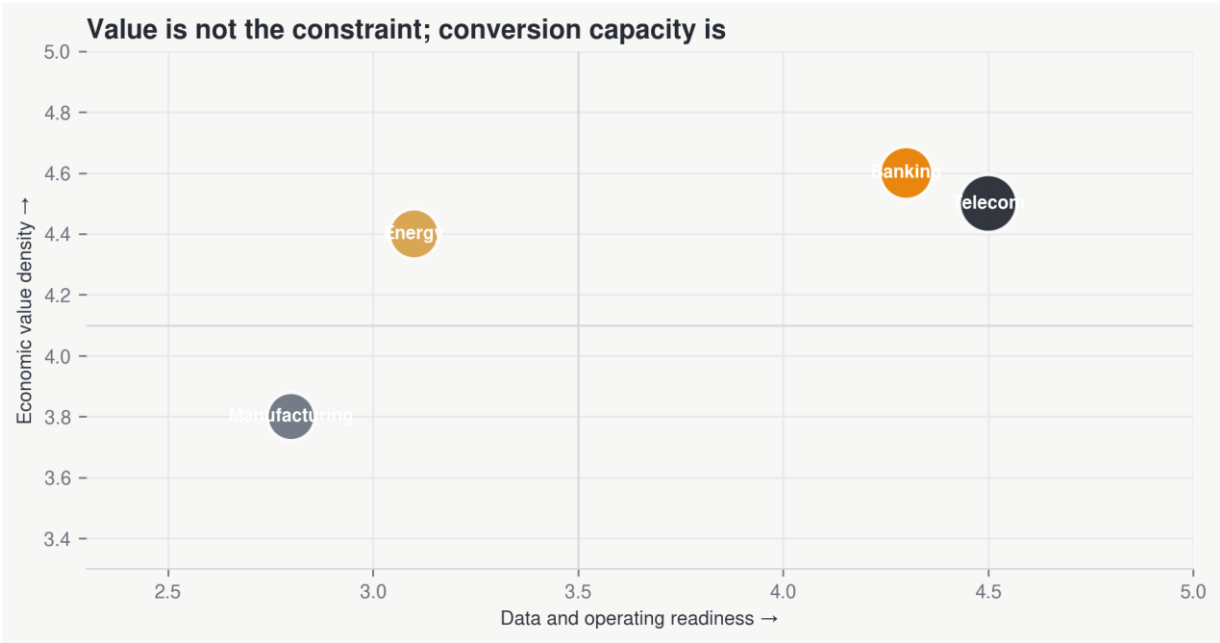


Exhibit 3. Operating readiness versus economic value density
Source: Renalytica analyst assessment, 1–5 scale. Bubble size is visual emphasis, not market size.

Sector	First portfolio bet	Do not lead with
Banking	service, fraud, compliance and collections	unexplainable autonomous credit decisions
Telecom	network, care, energy and churn	unbounded autonomous network control
Energy	maintenance, forecasting and anomaly detection	closed-loop control before shadow evidence
Manufacturing	quality, maintenance, yield and energy	enterprise-wide transformation without instrumentation

PORTFOLIO LOGIC

Allocate capital across horizons: 60–70% to proven assist and decide use cases, 20–30% to reusable data and control foundations, and no more than 10–15% to high-autonomy options until evidence improves.

Allocation is a Renalytica portfolio heuristic, not a universal benchmark. Adjust for sector, control maturity and strategic intent.

02

Scaling the investment

The economics change when pilots become a production portfolio.

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INVESTMENT ECONOMICS

The model is rarely the largest cost

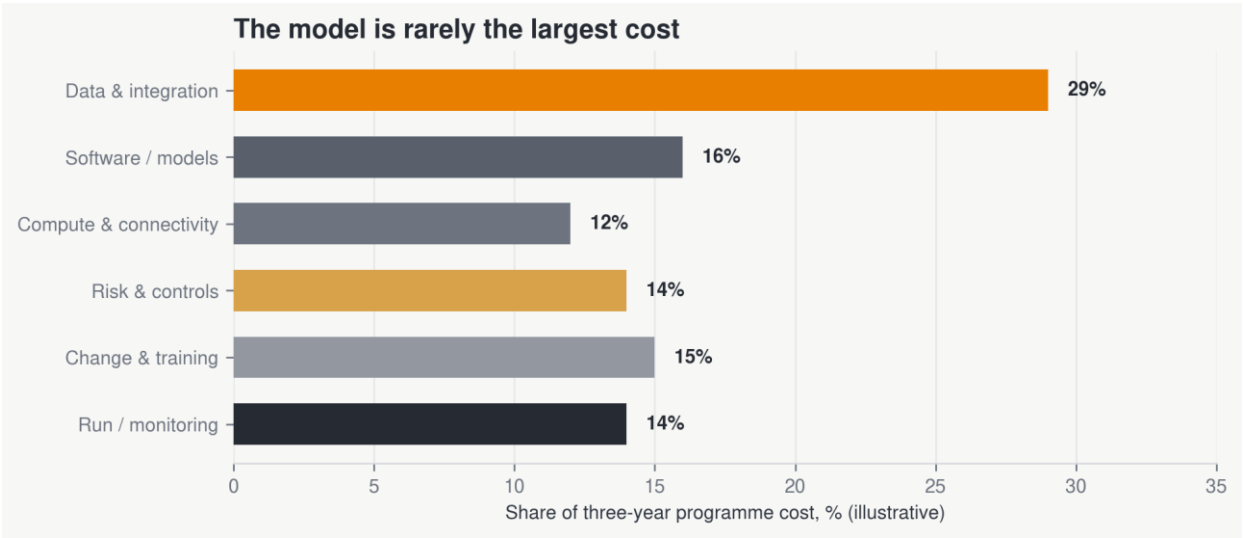


Exhibit 4. Illustrative three-year AI programme cost stack
Source: Renalytica illustrative model. Percentages are a decision heuristic and will vary by architecture and sector.

Direct model and software access can look inexpensive in a pilot. Production economics change when the organisation adds data pipelines, identity, integration, evaluation, monitoring, human review, security, process redesign and user adoption. The marginal cost of the second use case falls only if these capabilities are genuinely reusable.

Cost test	Question for investment committee
Fixed versus variable	Which costs recur with every model call, user, site or transaction?
Reuse	Which data, controls and integration components serve the next three use cases?
Fallback	What capacity must remain when the model is unavailable or uncertain?
Remediation	Who pays for drift, poor outputs, customer redress or operational recovery?

INVESTMENT ECONOMICS

Payback is a range, not a promise

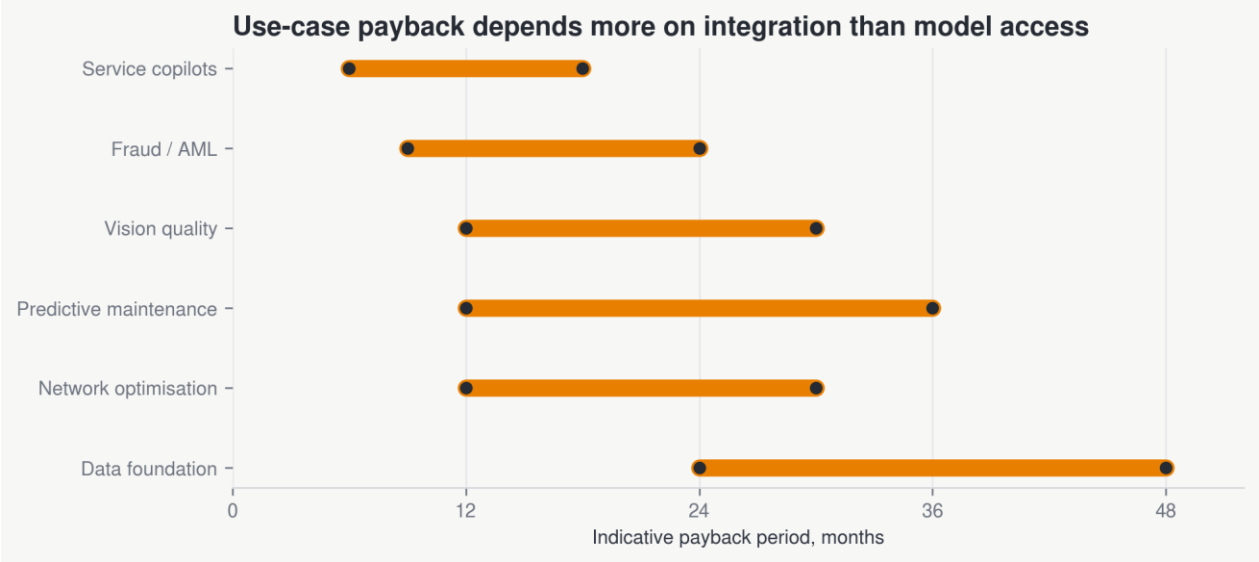


Exhibit 5. Indicative payback bands for selected AI investments
Source: Renalytica analyst ranges. Payback starts at production approval and assumes a measurable baseline; it is not a Nigeria market benchmark.

INTERPRETATION	
Shorter bands usually belong to assistive use cases that sit on existing digital workflows. Longer bands reflect data foundations, physical assets, change complexity or higher assurance requirements.	
Gate	Approve only if...
Pilot	baseline, owner, users, data and decision rights are explicit
Production	quality, security, privacy, fallback and unit economics pass
Scale	realised value survives adoption and run cost
Renew	performance remains above a simpler rule or non-AI alternative

OPERATING MODEL

Build, buy and partner around the source of advantage

Component	Default posture	Why
Foundation model	buy / access	capability changes quickly; switching and evaluation matter
Sector workflow	buy then configure	speed where process is standard; protect exit rights
Proprietary data layer	build / govern	contains context, history and differentiated signals
Integration and controls	own architecture	determines resilience, auditability and reuse
Specialist delivery	partner selectively	fills scarce skills; avoid permanent dependency

The minimum production team

BUSINESS SPINE Value owner, workflow owner, frontline users, finance partner and risk owner jointly define the baseline and intervention.	TECHNICAL SPINE Data engineering, model/product engineering, security, privacy, evaluation and platform operations keep the capability reliable.
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Three procurement clauses that protect value

- Data-use boundaries, retention, location and deletion obligations.
- Performance, audit, incident, model-change and subprocessor transparency.
- Portability of prompts, evaluations, integrations and business records on exit.

BOARD QUESTION If the vendor disappeared tomorrow, what operating knowledge, data, evidence and controls would remain inside the organisation?
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GOVERNANCE AND WORKFORCE

Trust is part of the return calculation

Nigeria’s data-protection framework gives individuals rights relating to automated decision-making, while NITDA’s responsible-AI messaging stresses trust, accountability and transparency. Governance therefore affects both downside risk and adoption: staff and customers are more likely to use systems whose role, limitations and escalation path are clear.

Risk	Economic transmission	Minimum control
Wrong or fabricated output	rework, loss, customer harm	grounding, confidence threshold, review
Bias / unfair treatment	exclusion, redress, conduct risk	cohort evaluation, override, appeal
Privacy / leakage	regulatory and trust loss	purpose limits, access, minimisation
Cyber / model attack	fraud, outage, unsafe action	threat modelling, isolation, monitoring
Workforce rejection	benefits remain theoretical	job redesign, training, feedback

Work should be decomposed—not counted away

TASKS LIKELY TO COMPRESS

Search, summarisation, first drafts, classification, routine triage and basic monitoring.

TASKS LIKELY TO EXPAND

Exception handling, judgement, assurance, data stewardship, customer explanation and process redesign.

PEOPLE METRIC

Track adoption, override, rework, escalation and quality by role. Hours theoretically saved are not value until capacity is redeployed or service improves.

03

Outlook and action

The 2030 spread will be created by operating-model choices made now.

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OUTLOOK TO 2030

Three scenarios for realised AI value

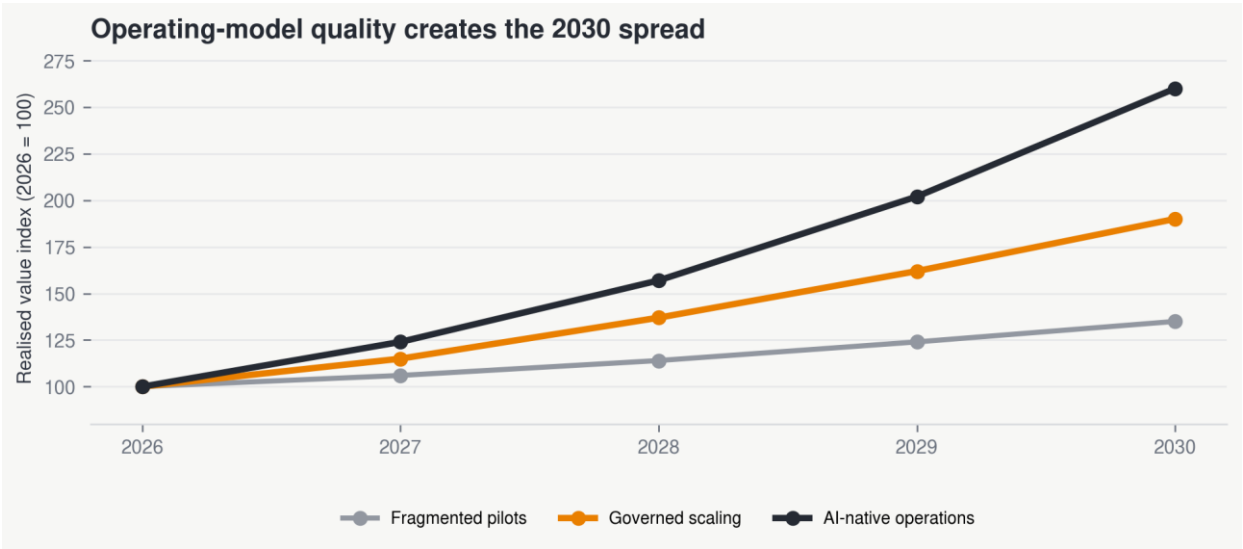


Exhibit 6. Illustrative realised-value index to 2030
Source: Renalytica scenario model. Index is not a monetary forecast and does not represent national GDP impact.

Scenario	What happens	Signpost
Fragmented pilots	many tools; low workflow integration; value leakage	pilot count rises faster than production use
Governed scaling	shared data and controls; portfolio discipline	reuse and measured value improve
AI-native operations	decision workflows redesigned; bounded autonomy	unit cost and cycle time structurally reset

SCENARIO INSIGHT

The difference is not access to better models. It is the cumulative effect of data reliability, workflow redesign, control maturity, adoption and reuse.

ACTION AGENDA

A 90-day route from AI ambition to investable portfolio

01 DAYS 1–15 inventory decisions	02 DAYS 16–30 baseline value	03 DAYS 31–50 rank and design	04 DAYS 51–75 shadow / pilot	05 DAYS 76–90 fund the portfolio
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Action	Output	Executive owner
Map 20–30 material decisions	decision inventory with frequency, loss and data	COO / sector operations
Select 3–5 use cases	value tree, baseline, intervention and counterfactual	CFO + business owners
Define production controls	risk tier, evaluation, privacy, security and fallback	CRO / CISO / DPO
Choose architecture and vendors	reusable data/integration plan and exit rights	CIO / CTO
Launch shadow or bounded pilots	weekly quality, adoption and economics review	product + frontline owner
Set stop / scale gates	funding released against realised evidence	investment committee

RENALYTICA RECOMMENDATION

Create one cross-functional AI value office with a finite mandate: establish the portfolio method, shared controls and evidence standard—then push value ownership into the operating businesses.

METHODOLOGY

How Renalytica formed the view

The report is based on desk research available up to 25 August 2026. Public sources were used to establish sector operating context, named company cases and regulatory direction. Renalytica then applied a common economic framework across the four sectors.

Assessment dimension	Interpretation
Data readiness	availability, quality and accessibility of decision-relevant data
Value density	economic value of improving a bounded decision
Speed to value	time from pilot to measurable operating effect
Scale repeatability	ability to reuse a solution across users, assets or sites
Control burden	cost and complexity of safety, conduct, privacy and audit controls

Limitations

Company-reported adoption indicators may reflect successful cases and are not sector samples. Macroeconomic indicators establish operating scale or pressure but do not measure AI adoption. All scores, value mixes, payback bands, portfolio allocations and scenarios are judgement-based and should be recalibrated with organisation-specific data before investment.

Selected primary and official sources

1. [NITDA — National Centre for AI and Robotics](#)
2. [NITDA — Responsible AI adoption, 2026](#)
3. [Nigeria Data Protection Commission](#)
4. [Central Bank of Nigeria — reforms](#)
5. [Nigerian Communications Commission — industry statistics](#)
6. [Nigerian Electricity Regulatory Commission — operating factsheets](#)

SOURCES AND CONCLUSION

What the evidence supports

- 7. [UBA — Leo adoption and digital banking disclosure](#)
- 8. [FirstBank — Fibani virtual assistant](#)
- 9. [MTN Nigeria — data-centre investment](#)
- 10. [Airtel Africa — data-centre infrastructure](#)
- 11. [NNPC RTI — robotic process automation](#)
- 12. [Dangote Cement — results and presentations](#)
- 13. [NERC — AI and operational compliance discussion](#)

Renalytica conclusion

THE ECONOMICS OF ADOPTION

Nigeria’s leading AI adopters will not be the organisations with the most pilots. They will be the ones that repeatedly select material decisions, establish a baseline, embed models in accountable workflows, price the full cost of production and stop weak use cases early.

Banking and telecom should move fastest where outcomes are frequent and reversible. Energy and manufacturing should move most deliberately where AI touches physical assets, safety or production control—but can still build strong returns through bounded maintenance, quality, forecasting and optimisation use cases. Across all four sectors, data, integration, change and governance are not support activities. They are the conversion engine that turns model capability into economic value.

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FROM AI ACTIVITY TO ECONOMIC VALUE

Market intelligence • Decision economics • Strategic analysis

ABOUT RENALYTICA

Renalytica produces evidence-led market intelligence focused on Nigeria's digital economy, financial infrastructure and emerging industries. Our work is designed to support better commercial, investment and policy decisions.

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